Homework - II

Statistics III : Multivariate Data and Regression

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Question 1

Consider a mutiple linear regression model with p predictors and an intercept. Assume that the errors are independent with mean zero and variance σ^2. Let $\hat{y}_i$ be the fitted value of the response Y for the i-th observation with $i = 1, \ldots, n$.

(a) Find the value of $\sum_{i=1}^{n} \text{Var}(\hat{y}_i)$ in terms of n, p and σ^2. Show all your derivation.

(b) State any additional assumptions that you may need.

(c) What does this result say about the impact of including more predictors in the model?

Solution 1

(a) Consider the following MLR model

$$Y = X\beta + \varepsilon$$

such that $E(\varepsilon) = 0$ and $Var(\varepsilon) = \sigma^2 I_n$. We know that the fitted values are given by $\hat{Y} = X\hat{\beta}_{LS}$ where $\hat{\beta}_{LS} = (X^TX)^{-1}X^TY$. Thus, we can write $\hat{Y}$ as

$$\hat{Y} = X(X^TX)^{-1}X^TY = HY$$

where $H = X(X^TX)^{-1}X^T$ is an $n \times n$ matrix. Consider the variance of the fitted values below

$$\begin{aligned} Var(\hat{Y}) &= Var(HY) \\ &= HVar(Y)H^T \\ &= H(\sigma^2 T_n)H^T \\ &= \sigma^2 HH^T = \sigma^2 H \qquad (\because H = H^T \text{ and } H^2 = H) \end{aligned}$$

We know that the $Var(\hat{y}_i)$ is simply the i-th diagonal element of $\sigma^2 H$. Thus, we have the following

$$\begin{aligned} \sum_{i=1}^{n} Var(\hat{y}_i) &= tr(Var(\hat{(Y)})) \\ &= \sigma^2 tr(H) \\ &= \sigma^2 tr(X(X^TX)^{-1}X^T) \\ &= \sigma^2 tr((X^TX)^{-1}X^TX) = \sigma^2 tr(I_{p+1}) \qquad (\because tr(AB) = tr(BA)) \\ &= \sigma^2(p+1) \end{aligned}$$

Thus, we have the following

$$\boxed{\sum_{i=1}^{n} Var(\hat{y}_i) = \sigma^2(p+1)}$$

(b) We have additionally assumed that X is fixed and has full column rank, i.e., $\text{rank}(X) = p+1$ for $(X^TX)^{-1}$ to exist. Also, we have assumed homoscedasticity and $Cov(\varepsilon_i, \varepsilon_j) = \sigma^2\delta_{ij}$ where δ_{ij} is the kronecker's delta

(c) We can conclude that increasing the number of predictor variables i.e., increasing p will linearly increase the variance of of fitted values since the sum of variances of the fitted values is directly proportional to $(p+1)$

Question 2

Consider simple linear regression and the problem of testing $\beta_1 = 0$. We can do this in two ways:

A. Use a t-test based on $\hat{\beta}_1 \sim \mathcal{N}(\beta_1, \sigma^2/S_{XX})$.

B. Use the F test in the ANOVA.

Answer the following questions in this context.

(a) Write down an estimator for σ^2 that is independent of $\hat{\beta}_1$.

(b) What is the distribution of the estimator in part (a)?

(c) Write down the t-statistic mentioned in (A) and its distribution.

(d) Show that the F-statistic in (B) is $\frac{S_{YY}-\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2/(n-2)}$.

(e) Show that the numerator in the F-statistic equals S_{XY}^2/S_{XX}.

(f) Hence show that the F-statistic in (B) is the square of the t-statistic in (A).

(g) Conclude that the two tests (A) and (B) are equivalent by expressing the t and F distributions in terms of independent normals.

Solution 2

(a) Consider the following

$$\boxed{s^2 = \frac{1}{n-2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{SSRes}{n-2}}$$

We can see that s^2 is independent of $\hat{\beta}_1$. We know know that $E(s^2) = \sigma^2$, i.e., s^2 is an unbiased estimator for σ^2 which also happens to be independent of $\hat{\beta}_1$.

(b) We'll proceed by making an assumption that $\varepsilon_i \sim N(0, \sigma^2)\ \forall 1 \leq i \leq n$. Thus, we know that

$$\sum_{i=1}^{n}(y_i - \hat{y}_i) \sim \sigma^2\chi^2_{(n-2)}$$

This implies that

$$s^2 \sim \frac{\sigma^2}{n-2}\chi^2_{(n-2)}$$

or alternativly

$$\boxed{\frac{(n-2)s^2}{\sigma^2} \sim \chi^2_{(n-2)}}$$

(c) We are given that

$$\hat{\beta}_1 \sim N(\beta_1, \sigma^2/S_{XX})$$

Thus, clearly

$$\sqrt{S_{XX}}\frac{\hat{\beta}_1 - \beta_1}{\sigma} \sim N(0,1)$$

Also, as proved in part (b),

$$\frac{(n-2)s^2}{\sigma^2} \sim \chi^2_{(n-2)}$$

This implies

$$\sqrt{S_{XX}}\frac{\hat{\beta}_1}{\sigma} \times \frac{\sigma(n-2)}{s(n-2)} = \sqrt{S_{XX}}\frac{\hat{\beta}_1}{s} \sim t_{n-2} \qquad (\because \text{we are testing for } \beta_1 = 0)$$

$\iff$ $\hat{\beta}_1$ and s^2 are independent. Consider for multivariate case, we have the following

$$\hat{\beta} = (X^TX)^{-1}X^T(X\beta + \varepsilon) = \beta + (X^TX)^{-1}X^T\varepsilon$$

and

$$\begin{aligned} Y - X\hat{\beta} &= (X\beta + \varepsilon) - X(X^TX)^{-1}X^T(X\beta + \varepsilon) \\ &= X\beta + \varepsilon - X\beta - X(X^TX)^{-1}X^T\varepsilon = (I - X(X^TX)^{-1}X^T)\varepsilon \end{aligned}$$

Thus, consider the following

$$\begin{aligned} Cov(\hat{\beta}, Y - X\hat{\beta}) &= Cov(\beta + (X^TX)^{-1}X^T\varepsilon, (I - X(X^TX)^{-1}X^T)\varepsilon) \\ &= \underbrace{Cov(\beta, (I - X(X^TX)^{-1}X^T)\varepsilon)}_{=0} + (X^TX)^{-1}X^T(\sigma^2 I)(I - X(X^TX)^{-1}X^T) \\ &= \sigma^2((X^TX)^{-1}X^T - (X^TX)^{-1}X^T) = 0 \end{aligned}$$

This implies $\hat{\beta}$ and $Y - X\hat{\beta}$ are independent. Thus, we firmly state our final result that

$$\boxed{\sqrt{S_{XX}}\frac{\hat{\beta}_1}{s} \sim t_{n-2}}$$

(d) We know that the $F-$statistic is defined as following

$$F = \frac{MSReg}{MSRes} = \frac{SSReg/1}{SSRes/(n-2)}$$

Since $SST = SSReg + SSRes$ such that $SST = S_{YY}$, this implies $SSReg = S_{YY} - \sum_{i=1}^n (y_i - \hat{y}_i)^2$. Thus, we have the following result

$$\boxed{F = \frac{S_{YY} - \sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \hat{y}_i)^2/(n-2)}}$$

(e) We know that $SSReg = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$. Thus, we have the following

$$\hat{y}_i = \bar{y} + \hat{\beta}_1(x_i - \bar{x})$$

Hence, we have the following expression for $SSReg$

$$\begin{aligned} SSReg &= \sum_{i=1}^{n} \left(\bar{y} + \hat{\beta}_1(x_i - \bar{x}) - \bar{y}\right)^2 \\ &= \sum_{i=1}^{n} \left(\hat{\beta}_1(x_i - \bar{x})\right)^2 \\ &= \hat{\beta}_1^2 \sum_{i=1}^{n} (x_i - \bar{x})^2 = \hat{\beta}_1^2 S_{XX}. \end{aligned}$$

We also know that $\hat{\beta}_1 = \frac{S_{XY}}{S_{XX}}$, thus we have the following

$$SSReg = \left(\frac{S_{XY}}{S_{XX}}\right)^2 S_{XX} = \frac{S_{XY}^2}{S_{XX}^2} S_{XX} = \frac{S_{XY}^2}{S_{XX}}.$$

Thus, the numerator equals S_{XY}^2/S_{XX}.

(f) Consider the square of $t-$statistic. We have the following

$$\begin{aligned} t^2 &= \left(\frac{\hat{\beta}_1}{s/\sqrt{S_{XX}}}\right)^2 \\ &= \frac{\hat{\beta}_1^2 S_{XX}}{s^2} \\ &= \frac{SSReg}{MSRes} \end{aligned}$$

We already know from previous parts that $\frac{SSReg}{MSRes}$ is exactly the formula for the F-statistic. Therefore,

$$\boxed{t^2 = F}$$

(g) A t-distribution with k degrees of freedom is defined as the ratio of a standard normal random variable Z to the square root of an independent Chi-square random variable U divided by its degrees of freedom: $t_k = \frac{Z}{\sqrt{U/k}}$.

Squaring this gives $t_k^2 = \frac{Z^2}{U/k}$. Since $Z \sim \mathcal{N}(0,1)$, its square Z^2 follows a Chi-square distribution with 1 degree of freedom (χ_1^2). An F-distribution F_{ν_1,ν_2} is defined as the ratio of two independent Chi-square variables, each divided by its respective degrees of freedom. Thus:

$$t_{n-2}^2 = \frac{\chi_1^2/1}{\chi_{n-2}^2/(n-2)} \sim F_{1,n-2}$$

Since Z (related to $\hat{\beta}_1$) and U (related to s^2) are independent normal-based statistics, the squared test statistic t^2 identically follows the F-distribution. Thus, the two tests are strictly equivalent for testing $H_0 : \beta_1 = 0$.

Question 3

Consider the data in the file `seatpos.xlsx` **Data description:** Car drivers like to adjust the seat position for their own comfort. Car designers would find it helpful to know where different drivers will position the seat depending on their size and age. Researchers at the HuMoSim laboratory at the University of Michigan collected data on 38 drivers.

(a) Fit a multiple linear regression of `hipcenter` on all other variables.

(b) Comment on the significance of the coefficients and the R^2.

(c) What do you suspect about collinearity from your comment? Compute the correlation matrix of the predictors to check if your suspicion is correct. Explain your conclusion.

(d) Fit simple linear regression models of `hipcenter` on each of the predictors separately. Which predictor gives the highest R^2? Is it very different from the R^2 you obtained from part (b)? Is the regression coefficient significant?

(e) Summarize your findings.

Solution 3

(a) Consider the following MLR fit of `hipcenter` on all seven predictors.

Listing 1: Multiple linear regression of hipcenter on all other variables

```
library(readxl)
seatpos <- read_excel("seatpos.xlsx")
seatpos <- na.omit(seatpos)

mlr_model <- lm(hipcenter ~ Age + Weight + HtShoes +
    Ht + Seated + Arm + Thigh + Leg, data = seatpos)
summary(mlr_model)
```

(b) We get $R^2 = 0.6866$, so about 68.7% of the variance in `hipcenter` is explained. Despite this, none of the individual predictors are significant at $\alpha = 0.05$:

- Age: $p = 0.184$, Weight: $p = 0.937$, HtShoes: $p = 0.784$, Ht: $p = 0.953$
- Seated: $p = 0.888$, Arm: $p = 0.736$, Thigh: $p = 0.671$, Leg: $p = 0.182$

(c) A high overall R^2 with no significant individual coefficients points to multicollinearity: standard errors inflate and t-statistics shrink even when the fit is good overall. Consider the correlation matrix.

Listing 2: Correlation matrix of predictors

```
predictors <- subset(seatpos, select = -hipcenter)
cor_matrix <- cor(predictors)
print(round(cor_matrix, 3))
```

	Age	Weight	HtShoes	Ht	Seated	Arm	Thigh	Leg
Age	1.000	0.081	-0.079	-0.090	-0.170	0.360	0.091	-0.042
Weight	0.081	1.000	0.828	0.829	0.776	0.698	0.573	0.784
HtShoes	-0.079	0.828	1.000	0.998	0.930	0.752	0.725	0.908
Ht	-0.090	0.829	0.998	1.000	0.928	0.752	0.735	0.910
Seated	-0.170	0.776	0.930	0.928	1.000	0.625	0.607	0.812
Arm	0.360	0.698	0.752	0.752	0.625	1.000	0.671	0.754
Thigh	0.091	0.573	0.725	0.735	0.607	0.671	1.000	0.650
Leg	-0.042	0.784	0.908	0.910	0.812	0.754	0.650	1.000

Table 1: Pearson correlation matrix of the predictor variables.

We know `Ht` and `HtShoes` correlate at 0.998, and `Ht` is also strongly correlated with `Seated` (0.928) and `Leg` (0.910). This confirms severe multicollinearity among the size-related predictors.

(d) Listing 3: Simple linear regression for each predictor

```r
predictors_names <- names(predictors)
results <- data.frame(Predictor = character(), R_
    Squared = numeric(), P_Value = numeric())

for (var in predictors_names) {
formula <- as.formula(paste("hipcenter ~", var))
model <- lm(formula, data = seatpos)
summ <- summary(model)
results <- rbind(results, data.frame(
     Predictor = var,
     R_Squared = summ$r.squared,
     P_Value = coef(summ)[2, 4]
))
}
print(results)
```

Predictor	R^2	p-value
Age	0.0421	2.166×10^{-1}
Weight	0.4100	1.493×10^{-5}
HtShoes	0.6346	2.207×10^{-9}
Ht	**0.6383**	$\mathbf{1.831 \times 10^{-9}}$
Seated	0.5347	1.845×10^{-7}
Arm	0.3423	1.142×10^{-4}
Thigh	0.3495	9.290×10^{-5}
Leg	0.6196	4.587×10^{-9}

Table 2: Results of SLR models for each predictor.

`Ht` gives the highest R^2 (0.6383), close to the MLR value of 0.6866. Unlike in the MLR, its coefficient here is highly significant ($p = 1.831 \times 10^{-9}$).

(e) The predictors are strongly intercorrelated. Taller drivers tend to be heavier with longer arms, legs and seated height and hence fitting them jointly destabilizes the individual coefficients despite a good overall fit. A single predictor like Ht captures nearly the same explanatory power while staying significant and far easier to interpret.

Question 4

Consider the data in the file earthquake.xlsx
Data description: The earthquake data frame contains measurements of latitude, longitude, focal depth and magnitude for all earthquakes having magnitude greater than 5.8 between 1964 and 1985.
Produce the diagnostic plots of residuals for the multiple linear regression of magnitude using all other variables as predictors. Comment on the plots.

Solution 4

Consider the following MLR fit and diagnostic plots.

Listing 4: Diagnostic plots for earthquake regression

```
library(readxl)
earthquake <- read_excel("earthquake.xlsx")
earthquake <- na.omit(earthquake)

eq_model <- lm(magnitude ~ depth + latitude + longitude, data =
    earthquake)
summary(eq_model)

par(mfrow = c(2, 2))
plot(eq_model)
```

Before looking at the plots: $R^2 = 0.020$ and none of the predictors are significant (smallest p-value is 0.280, for depth). The model explains essentially none of the variation in magnitude.

Comment on the plots

- **Residuals vs Fitted:** Horizontal banding shows magnitude is discretely measured, not continuous.
- **Normal Q-Q:** Heavy right tail, residuals are non-normal, and the model underpredicts high-magnitude earthquakes.
- **Scale-Location:** Same discrete banding; variance otherwise looks roughly constant.
- **Residuals vs Leverage:** Point 1 is a low-leverage outlier with the largest residual; points 78, 87, 88 have high leverage but small residuals, so no single point is strongly influential.

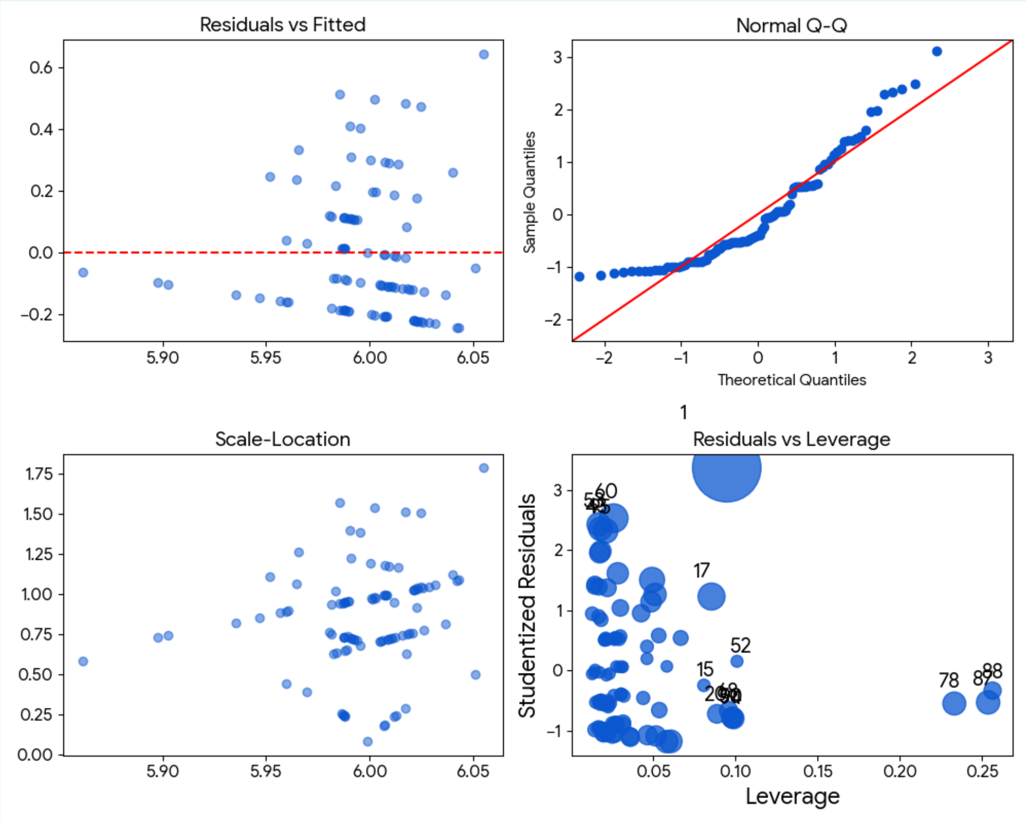

Figure 1: Standard diagnostic plots of residuals for the earthquake MLR model.

Overall: MLR is unsuitable here, not just due to discreteness and non-normal residuals, but because $R^2 \approx 0.02$ with no significant predictors means depth, latitude and longitude carry almost no linear information about magnitude.